Venkata Lakshmi Parimala Pasupuleti

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SUMMARY

Staff-track AI engineer with **9+ years in production enterprise SaaS** - now leading the design and delivery of **production-grade generative AI, agentic, retrieval-augmented generation, and evaluation systems** at Intuit. Define **technical strategy and architecture** for AI capabilities that reach millions of end users: agent orchestration (LangGraph), structured output validation, observability pipelines, and **responsible AI guardrails** (PII filtering, confidence-gated escalation, HITL controls). Built **AgentProbe** - an open-source multi-provider LLM evaluation platform and **Leasel** - an 8-agent production agentic system with full CI/CD evaluation harness (93.6% precision, 1,240+ test cases). At Entrata (100K+ MAU enterprise SaaS), **led cross-functional technical initiatives** across 120+ sprints: owned vendor integrations, established engineering standards, mentored engineers, and delivered distributed cloud-native services on AWS. Strong in **Python, LangGraph, FastAPI, distributed systems, AWS, Terraform, GitHub Actions, observability, and AI quality & evaluation methods**. M.S. Data Analytics Engineering, George Mason University (GPA 3.87, May 2026).

CORE COMPETENCIES

Generative AI & Agentic Systems (LangGraph) | Retrieval-Augmented Generation (RAG) | AI Evaluation Methods & LLMOps CI/CD

Technical Strategy & Architecture | Distributed Systems & Cloud-Native Services | Observability / Monitoring / Alerting

Responsible AI & AI Security Controls | Python / FastAPI / SQL / REST APIs | AWS (S3, EC2, RDS, Lambda, SQS, IAM)

Docker / Kubernetes / Terraform / GitHub Actions | Cross-functional Technical Leadership | Engineering Standards & Mentorship

EXPERIENCE

Intuit

Jun 2026 -- Present

Software Engineer - Agentic AI & Technical Leadership - Virginia, USA

- **Define technical strategy and architecture** for production-grade generative AI and agentic systems: LangGraph orchestration, RAG pipelines, structured output validation, and cloud-native AI services that scale across financial workflows for millions of users.
- Lead **AI evaluation and measurement** infrastructure: LLM-as-judge + golden dataset CI/CD pipelines (GitHub Actions), regression testing across model updates, and observability dashboards for agent system health - establishing engineering standards for AI quality at the organization level.
- Implement **responsible AI and security controls**: prompt injection safeguards, PII filtering/redaction, confidence-gated HITL escalation, and resiliency patterns - ensuring AI capabilities meet regulated financial data requirements.
- Communicate technical decisions and recommendations to **engineering leaders and business stakeholders**; evaluate emerging AI models and frameworks through rigorous experimentation; provide technical mentorship across teams.

Entrata - Multifamily & Affordable Housing SaaS

Oct 2020 -- Jul 2024

Senior Software Engineer - 100K+ monthly active users · Enterprise multi-tenant platform

- **Led complex, cross-functional technical initiatives** across 120+ Agile sprints: primary technical contact for HUD and Tax Credit federal vendor integrations - scoped requirements, resolved production incidents, and drove initiatives from discovery through go-live without handoffs.
- Defined and enforced **engineering standards** across distributed cloud-native services on AWS (S3, EC2, RDS, Lambda, SQS): Terraform IaC, GitHub Actions CI/CD, Docker microservices, observability (logging, alerting, monitoring), and incident response discipline.
- Built **production distributed systems**: RabbitMQ async processing for rent updates across thousands of tenant records (idempotency, retry logic); Python/PySpark data pipelines processing 1.6M+ records; internal automation eliminating 80% of manual operational work.
- Resolved **ambiguous technical problems** end-to-end: root cause analysis across stored procedures, distributed data pipelines, and integrations; wrote RCAs for non-technical stakeholders; mentored junior engineers; reviewed 300+ pull requests.

Entrata India

Sep 2018 -- Oct 2020

Junior Software Engineer - Pune, India

- Built production Python/Java/PostgreSQL modules for enterprise housing SaaS; delivered large-scale data pipelines (1.6M records) with validation and reconciliation.

PROJECTS

Leasel - Production Agentic AI System

Python · LangGraph · FastAPI · RAG · HITL · Responsible AI · LLMOps

Production-grade generative AI system for enterprise document intelligence: 8-agent LangGraph orchestration with real conditional edges, retrieval-augmented generation (hybrid TF-IDF + semantic reranking), structured output validation (Pydantic), and **responsible AI controls** - confidence-gated HITL escalation and hard routing rules that prevent LLMs from making regulated eligibility or financial decisions autonomously. Full **AI evaluation harness** integrated into CI/CD: LLM-as-judge + golden datasets, 93.6% precision, 87.9% F1 across 1,240 test cases.

github.com/venkatalakshmiparimala/Lease_complaince_rag

AgentProbe - Open-Source LLM Evaluation Platform

Python · FastAPI · LangGraph · 5 LLM Providers · Evaluation Methods · CI/CD

15-endpoint FastAPI platform integrating five LLM providers (Anthropic, OpenAI, Azure OpenAI, Groq, Ollama) with **standardized AI evaluation methods**: factual accuracy, hallucination rate, retrieval quality, completeness, and compliance scoring. **LLMOps CI/CD pipeline** (GitHub Actions) with automated regression tests against golden datasets - the same evaluation infrastructure needed to maintain AI quality as models and features evolve. Demonstrates systematic **experimentation and measurement** across provider and model updates.

github.com/venkatalakshmiparimala/AgentProbe

EDUCATION

M.S. Data Analytics Engineering — George Mason University, Virginia — May 2026 | GPA 3.87

B.Tech Computer Science & Engineering — JNTU Anantapur — 2017